Your portfolio.
Your workspace.

Deployment and maintenance guide - updated 8 October 2026. The portfolio, administrator dashboard, Portainer, and Fieldnotes share one Oracle instance and one HTTPS hostname.

Resource	Address / purpose
Public portfolio	https://rithvik.ddns.net/
Administrator dashboard	/admin/ - launch your applications
Shared login	/auth/ - Authelia, existing rithvik account
Portainer	/portainer/ - Docker container management
Fieldnotes editor	/admin/notes/ - private writing workspace
Public writing	/blog/ and /learn/ - published entries only

The deployed stack

Oracle IP: **144.24.135.201**. SSH user: **opc**. Repository: **/opt/portfolio**, server branch **main**. No-IP resolves the hostname; Caddy provides HTTPS and routes requests. Only Caddy publishes web ports.

Four containers run the stack: app, caddy, auth, and portainer. Fieldnotes runs inside the existing Node app, so it adds no extra container or database service. Private writing persists outside the image.

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This is a setup record, not a continuously monitored status page. Credentials and private content are deliberately excluded.

Docker files, explained.

File	What it does
Dockerfile	Builds the Node app image. Generates the public pages, checks links, and runs tests.
compose.yaml	Local public-site preview: one container, host localhost:8080 to internal port 4173.
compose.production.yaml	Oracle stack: app, Caddy, Authelia, Portainer, private networks and persistent storage.
compose.ip.yaml	Legacy temporary HTTP preview on a public IP. Do not run beside production.
.dockerignore	Allows only required build inputs. Private data, credentials and Git history stay out.
deploy/Caddyfile.docker	Active container routing and HTTPS configuration.
deploy/Caddyfile and portfolio.service	Alternative host/systemd templates; not used by the Docker stack.

A **Dockerfile creates an image**. **Compose runs the services** and describes ports, settings, networks and storage. Build makes an image; up creates or updates containers; up --build does both. Restart alone does not rebuild changed code.

Local Docker Desktop preview

```
docker compose config --quiet
docker compose up --build -d --wait
docker compose ps
```

Open **http://localhost:8080**. Port 4173 is internal to the container. Local Compose does not start shared login, and explicitly disables writing storage; public writing collections are empty. Use production for the authenticated editor.

```
docker compose logs --tail=100 app
docker compose down
```

A direct Node preview uses pnpm install, pnpm build, pnpm check, pnpm test, then pnpm dev at http://127.0.0.1:4173. Install dependencies first. Tests use fake email delivery.

One command to redeploy.

On your development computer: commit your changes, merge V2 into main, then push main to GitHub. Include generated HTML when the portfolio templates or data change. Do not commit .env files, .auth/, .portainer/, or .notes/.

Inside your Oracle SSH session:

```
bash /opt/portfolio_redeploy.sh
```

The shortcut already exists. It calls /opt/portfolio/portfolio_redeploy.sh. You do not need to create a new file or paste the script into the terminal.

What the script does

1. Locks against simultaneous deployments and fetches main.
2. Allows a fast-forward update, stopping on conflicting server changes.
3. Prepares private writing storage and Portainer/OIDC files.
4. Validates Compose, Authelia, and Caddy.
5. Builds the Node image, pages, links, and automated tests.
6. Recreates Authelia, reconciles Portainer SSO, and starts the stack.
7. Reloads Caddy and checks public pages and signed-out access gates.

The running site stays available during the build. The app briefly restarts when replaced. Authelia sessions expire on redeployment, so sign in again afterward. Writing, account data, Portainer settings and certificates persist.

Server copies and Git

Stages 1-3 were copied to Oracle before the local commit and merge. Merge the matching code and docs before a normal redeploy. The script stashes server copies only when they match origin/main exactly; differences stop the update. Reconcile the named files instead of discarding them.

Rebuild the existing server checkout

```
bash /opt/portfolio_redeploy.sh --no-pull
```

Use this after changing the saved server environment or when deliberately deploying the existing checkout. A Git push alone does not deploy. Editing .env locally does not edit Oracle .env.production.

One login, three tools.

Open **https://rithvik.ddns.net/admin/** and sign in as **rithvik**. The password is stored privately in Portfolio admin login.txt beside your Oracle key. The server bootstrap record is in .auth/bootstrap-login.txt. Store the credential in your password manager.

Shared session

Caddy verifies the Authelia session before allowing /admin/* or /portainer/*. It strips visitor-supplied identity headers. Node accepts the administrator identity only from the configured Caddy IP, for the configured user in the admins group.

Authelia sessions expire after 30 minutes of inactivity or eight hours total. Remember-me, TOTP, WebAuthn, and email password reset are not configured. Redeploying intentionally ends the in-memory sessions.

Portainer

Choose **Open Portainer**, then **Login with OAuth** if its login screen appears. It reuses the existing shared session. Open the **Oracle Docker** environment to inspect containers, logs, images and resource use.

Portainer has the Docker socket and can control the host containers. Prefer changing versioned Compose files for lasting configuration; a GUI-only change may be replaced by the next redeployment. A separate recovery admin exists in the private Portainer recovery login.txt file.

Sign out

The dashboard and editor sign-out link ends Authelia access. New Portainer requests then require login, but Portainer has a separate eight-hour token; use its own logout too when ending a management session. Close active console/WebSocket sessions as well.

Fieldnotes

Choose **Open Fieldnotes**. The editor is part of the Node app, directly using the verified shared identity. It needs no additional OAuth client or password. Public readers do not need an account.

Recovery procedures and the exact trust boundaries are in deploy/ADMIN.md and deploy/PORTAINER.md. Never paste secrets, cookies or resolved Compose configuration into public logs.

Write, save, then share.

Open **/admin/notes/**. Select **New entry**, choose Blog post or Learning note, add a title and optional summary, then write in Markdown. Both types start private.

The writing loop

Save draft (Ctrl+S / Cmd+S) saves to Oracle. Saving is manual: wait for confirmation before closing. **Preview** shows formatted Markdown. **Publish entry** asks for confirmation, then makes the entry visible to everyone on /blog/ or /learn/.

After publication, saving further edits changes only the private draft. Choose **Publish update** to replace the public version. The first publication date remains. Changing type and publishing moves the entry to the other collection, so the previous collection URL stops working.

Organize and recover

Add up to eight comma-separated tags. Search titles, tags or Markdown content. Filter by blog, notes, private drafts, published entries, or trash. Public searches only inspect published snapshots.

Unpublish removes the public version and keeps the draft. **Move to trash** also unpublishes. Select the Trash filter and **Restore draft** to recover a private draft. There is no permanent-delete button. Copies already downloaded by others cannot be recalled.

Markdown essentials

````
## Section heading
**Bold text** and *emphasis*
[Documentation](https://example.com)
- List item
- [ ] Task to complete

```sh
docker compose ps
```
````

Tables and fenced code blocks are supported. Images, uploaded attachments, embedded widgets, comments and scheduling are not included. The editor sanitizes generated HTML before preview and publication.

Unsaved writing lives only in the open tab. If a session expires, keep that tab open, sign in through Overview in another tab, then retry. A conflicting save is rejected; download or copy your text before reloading the newer version.

Writing survives a rebuild.

Private location	Purpose
.env.production	Oracle domain, SMTP and runtime settings
.auth/	Users, login keys and the matching auth database
.portainer/	Recovery/OAuth credentials and Portainer database
.notes/	Drafts, published snapshots, revisions and recoverable trash
Caddy named volumes	Certificates, ACME account and runtime state

Production mounts .notes/ at /writing in the Node container. The directory is mode 700 and files are mode 600, owned by UID/GID 1000. The rest of the app filesystem stays read-only. Git updates and image builds preserve these runtime files.

One process owns the writing store. Writes are serialized, revision-checked and atomically renamed. Limits: 250 entries including trash, 64 KiB UTF-8 Markdown per entry, 140-character titles, 320-character summaries and eight 32-character tags. Do not run multiple writers against the directory.

Export versus backup

Download Markdown saves current editor text, including unsaved changes. **Export library** downloads all saved entries as private JSON. There is no browser import in this version. A server restore uses the original .notes/ files.

Consistent private-state backup in SSH

```
cd /opt/portfolio
umask 077
backup=$(mktemp -d /home/opc/.config/portfolio-deploy/private-state-XXXXXXXX)
dc() { sudo docker compose --env-file .env.production \
  -f compose.production.yaml "$@"; }
dc stop app auth portainer
sudo tar -czf "$backup/private-state.tar.gz" \
  .env.production .auth .portainer .notes
sudo chmod 600 "$backup/private-state.tar.gz"
dc start app auth portainer
dc ps
```

If archiving fails, start the services again. The app is briefly unavailable during this backup. Also preserve matching source and Caddy volumes. Copy private backups to secure storage outside the VM; scheduled backups are not configured.

Caddy routes your sites.

The active configuration is **deploy/Caddyfile.docker**, mounted at /etc/caddy/site/Caddyfile.docker. Caddy receives HTTPS traffic, selects the hostname, and forwards the request to the correct private service.

Request	Handling
/auth/*	Preserve path; proxy to auth:9091
/admin/*	Check Authelia; forward verified identity to app:4173
/portainer/*	Check original URL, strip prefix, proxy to portainer:9000
/blog/* and /learn/*	Existing public fallback to app:4173
/locales/*/translation.json	Portainer compatibility redirect through its protected route

When adding another application

First check whether it supports a path such as /tool/ or requires its own hostname. Add its service to Compose and an appropriate network shared with Caddy. Leave host ports unpublished. Inside Caddy, use the service name and internal port, not localhost.

A public path on the existing domain needs no wildcard or DNS change. This example goes before the fallback handle and assumes the application supports the public base path:

```
redir /tool /tool/ 308
handle_path /tool/* {
    reverse_proxy tool:3000
}
```

handle preserves the path; **handle_path** strips the prefix. Assets and redirects must work with the chosen base URL. For a private app, also add its Authelia access rule and supported identity integration. Forward authentication does not automatically remove another application's own login.

Fieldnotes needs no new handler because the editor is already under /admin/* and public writing is handled by Node. A separate public hostname uses a top-level block:

```
tool.example.com {
    encode zstd gzip
    reverse_proxy tool:3000
}
```

These tool blocks are examples, not deployed services. Give the hostname a DNS record pointing to Oracle. Validate and reload Caddy, or use the redeploy script for combined changes.

Domains and first setup.

Why both DNS and the hostname matter

DNS finds the server IP. HTTPS still carries the hostname for TLS and HTTP routing. Caddy uses it to select the certificate and site. The app uses its public origin to validate contact/writing requests and article URLs; Authelia uses it for cookies and redirects.

This deployment redirects HTTP on the IP to the hostname. It is not configured for HTTPS directly on the IP. Keep the No-IP hostname renewed, and update DNS if the Oracle public IP changes. No Cloudflare tunnel, Worker, Pages deployment or automatic DNS updater is installed.

Move to a purchased domain

Point its DNS record to Oracle. Change DOMAIN in /opt/portfolio/.env.production to the hostname only, without a scheme or path. Keep private directories and certificate volumes, then run:

```
bash /opt/portfolio_redeploy.sh --no-pull
```

Compose updates the app origin and build URL, Authelia URLs/cookies, and Portainer trusted origin. The helper reconciles its OAuth callback and endpoints. Caddy obtains the new certificate. Sign in again on the new domain and check all three tools plus the two public writing collections.

A fresh Linux server

Install Docker Engine, Compose, Git, Python 3, OpenSSL, curl and flock. Clone main into /opt/portfolio. Use the deployment user UID/GID 1000 for the current Node storage setup, or deliberately adapt container and directory ownership. Set DNS and permit TCP 80/443 in Oracle and host firewalls; keep private app ports closed.

```
cd /opt/portfolio
cp -n deploy/production.env.example .env.production
chmod 600 .env.production
# Edit DOMAIN and private SMTP settings, then:
python3 scripts/setup-admin.py
bash portfolio_redeploy.sh --no-pull
```

The administrator initializer is one-time and refuses to overwrite existing auth data. The redeploy script prepares Portainer and writing storage. Docker builds Node inside the image. The /opt wrapper is specific to the current server; use the repository script on a fresh host.

Use provider SMTP/app credentials and an allowed sender. Health checks show configured delivery, not inbox receipt. No actual email is sent by automated tests.

Check health. Know the limits.

```
cd /opt/portfolio
sudo docker compose --env-file .env.production \
  -f compose.production.yaml ps
sudo docker compose --env-file .env.production \
  -f compose.production.yaml logs --tail=80 app caddy auth
curl -I https://rithvik.ddns.net/
curl -I https://rithvik.ddns.net/admin/notes/
curl https://rithvik.ddns.net/api/contact
free -m
sudo docker stats --no-stream
```

Public pages should return 200. Signed-out private paths redirect to login or return 401. A private article URL returns 404. Contact available:true means SMTP is configured, not that a message reached the inbox.

The 8 October memory incident

The website and SSH stalled even though Oracle reported Running. A reboot restored access. Persistent /var/log/messages recorded repeated host out-of-memory events killing dnf in dnf-makecache.service on 7 October at 23:31 UTC, and 8 October at 06:56 and 09:29 UTC.

During the last event, DNF held roughly 596 MiB resident memory and only 252 KiB of about 2.5 GiB swap remained free. This strongly suggests memory/swap exhaustion caused the outage; no live sample was possible while SSH failed. Some events preceded Portainer installation.

```
sudo systemctl disable --now dnf-makecache.timer
sudo systemctl stop dnf-makecache.service
```

This targeted mitigation is already applied. It disables background metadata caching, not package installation or upgrades. Manual DNF operations can still need substantial memory; plan them for a maintenance window. The host setting survives reboot but is not part of the redeploy script.

After resolving memory capacity, scheduled caching can be restored with `sudo systemctl enable --now dnf-makecache.timer`. The prior unit state is saved in the private dnf-cache-outage-MGfiPW0L recovery directory.

The VM has about 945 MB RAM. Container limits are ceilings, not reservations: app/Portainer 256 MiB each, Caddy/Authelia 192 MiB each. Builds and the OS also need memory. The unrelated failed mcelog service reports unsupported AMD hardware.

Recover carefully. Keep a copy.

Restore private state as a matched set

Stop the affected services first. Preserve the current data as a separate recovery point. Extract the verified archive into staging, then restore the matching private directories with owners and permissions intact. Replace the writing directory as a unit; overlaying an old backup could leave newer entries published.

Use a compatible source/image version. Keep the Authelia encryption key with its matching database and preserve paired Portainer OAuth credentials. Restart and verify both public content and private access. Do not edit writing JSON files while the app is running; it keeps an in-memory snapshot.

Current recovery points

Private parent: `/home/opc/.config/portfolio-deploy/`

Recovery point	Purpose
fieldnotes-stage3-DtZPMrmu/	Source and private state before Fieldnotes. Predates new writing data.
rithvik-portfolio:before-fieldnotes-stage3	Previous app image before the writing deployment.
portainer-stage2-6erYb3rT/	Earlier Portainer integration recovery point.
dnf-cache-outage-MGfiPW0L/	Saved background-cache unit configuration and incident documentation.

Repository reference map

README.md: current addresses and development workflow.
deploy/DOCKER.md: Docker, Oracle, routing, new domains and incident detail.
deploy/ADMIN.md: shared login, password recovery and trust boundaries.
deploy/PORTAINER.md: OAuth, container management and recovery account.
deploy/FIELDNOTES.md: writing workflow, storage, export and restoration.
deploy/README.md: alternative systemd setup.

Primary references

Caddy path handling - caddyserver.com
Authelia / Caddy integration - authelia.com
Portainer OAuth settings - docs.portainer.io
Marked Markdown parser - marked.js.org
sanitize-html - github.com/apostrophecms
DNF makecache - dnf.readthedocs.io